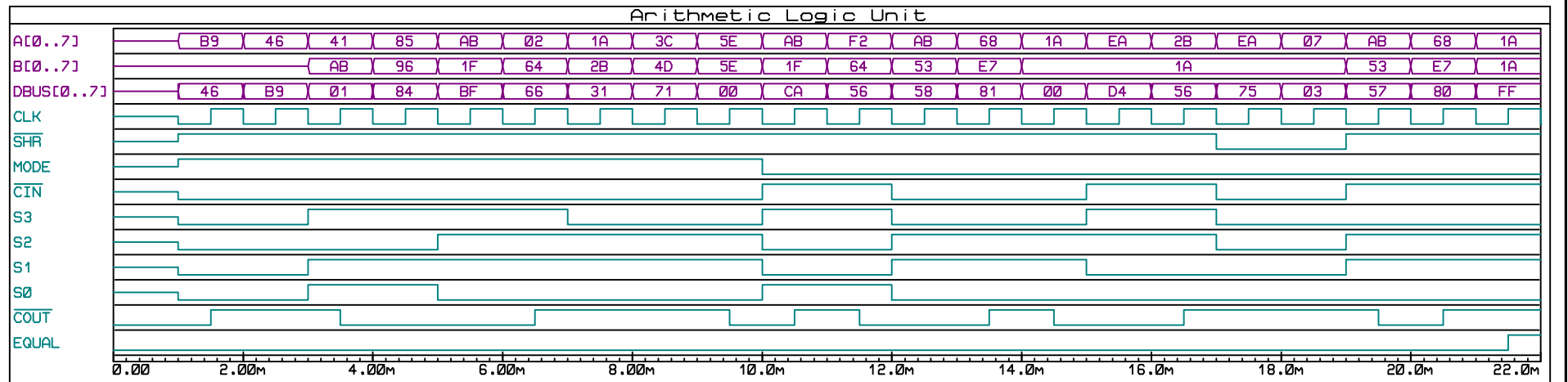
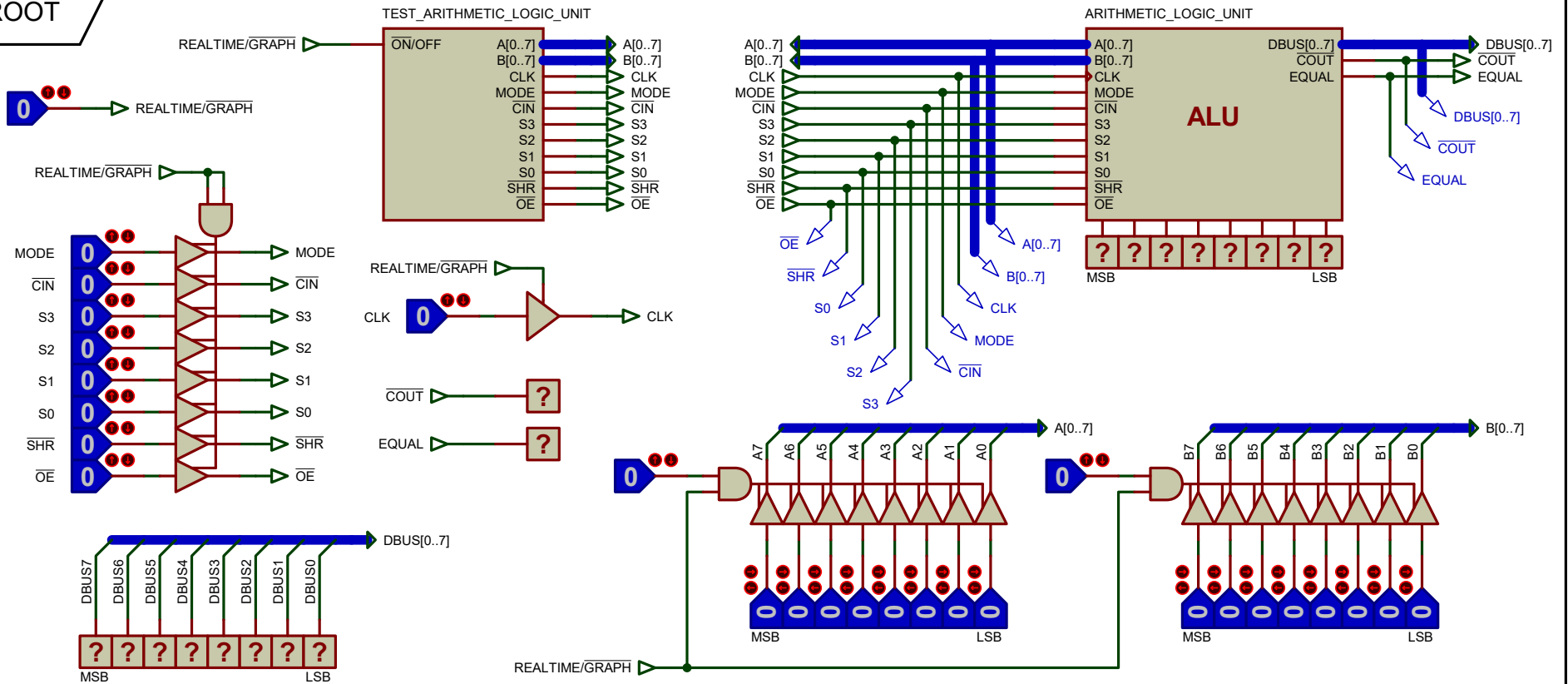
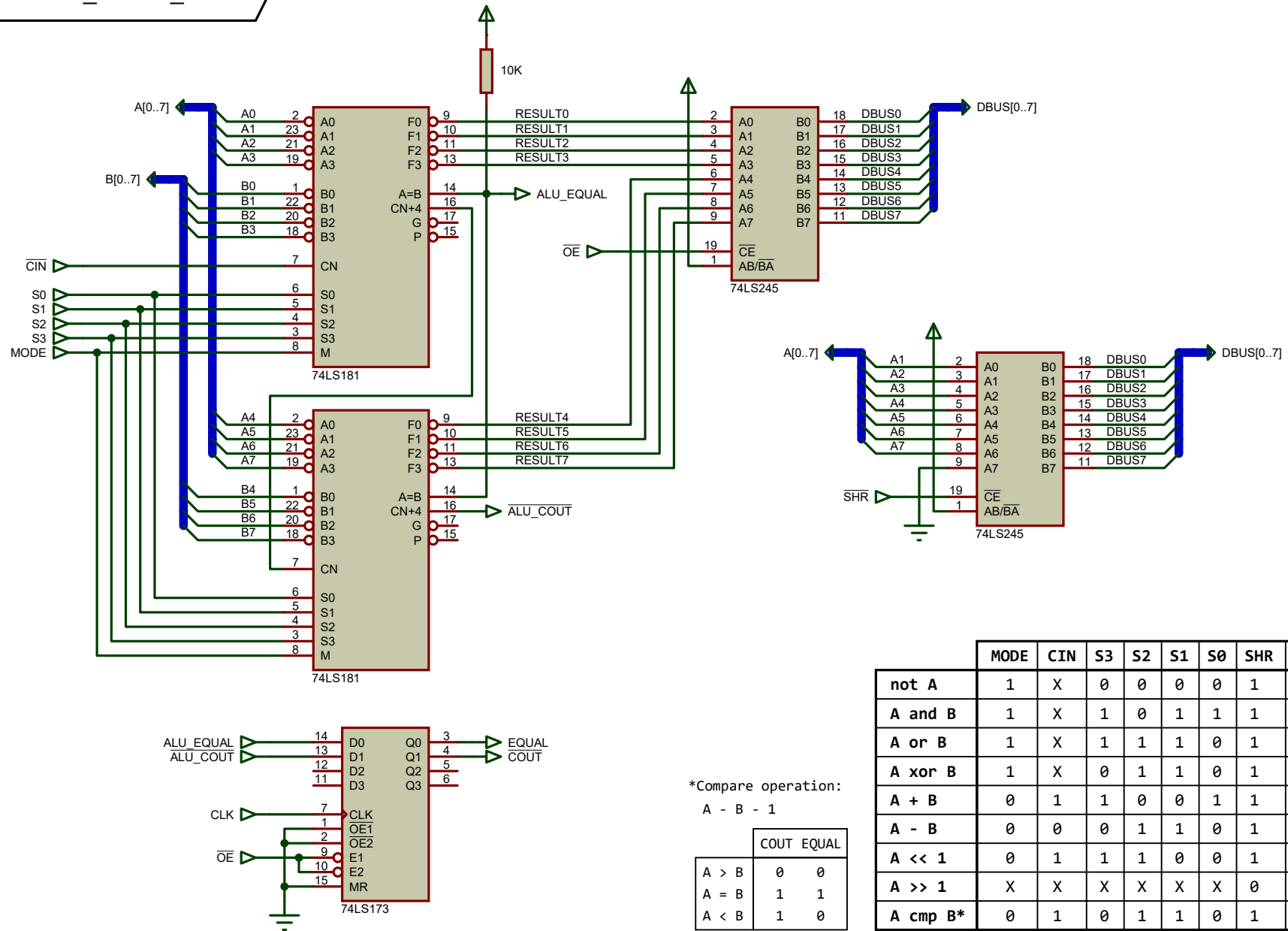


ROOT



ARITHMETIC_LOGIC_UNIT



*Compare operation:

$$A - B - 1$$

	COUT	EQUAL
A > B	0	0
A = B	1	1
A < B	1	0

	MODE	CIN	S3	S2	S1	S0	SHR	OE
not A	1	X	0	0	0	0	1	0
A and B	1	X	1	0	1	1	1	0
A or B	1	X	1	1	1	0	1	0
A xor B	1	X	0	1	1	0	1	0
A + B	0	1	1	0	0	1	1	0
A - B	0	0	0	1	1	0	1	0
A << 1	0	1	1	1	0	0	1	0
A >> 1	X	X	X	X	X	X	0	1
A cmp B*	0	1	0	1	1	0	1	0

TEST_ARITHMETIC_LOGIC_UNIT

*SCRIPT GENERATOR TEST_ARITHMETIC_LOGIC_UNIT

ALIAS ONOFF = \$ON\$/OFF
ALIAS CIN = \$CIN\$
ALIAS OE = \$OE\$
ALIAS SHR = \$SHR\$

PIN ONOFF
PIN A[0..7] = FLT
PIN B[0..7] = FLT
PIN MODE = FLT
PIN CIN = FLT
PIN S3 = FLT
PIN S2 = FLT
PIN S1 = FLT
PIN S0 = FLT
PIN SHR = FLT
PIN OE = FLT

INT OP_NOT = %10000010, OP_AND = %10101110, OP_OR = %10111010
INT OP_XOR = %10011010, OP_ADD = %01100110, OP_SUB = %00011010
INT OP_SHL = %01110010, OP_SHR = %00000001, OP_CMP = %01011010
INT OP

IF ONOFF = TRUE

A[0..7] = FLT
B[0..7] = FLT
MODE = FLT
CIN = FLT
S3 = FLT
S2 = FLT
S1 = FLT
S0 = FLT
SHR = FLT
OE = FLT

ELSE

SLEEP FOR 1m
OP = OP_NOT : A = \$B9 : GOSUB OPERATION // NOT
OP = OP_NOT : A = \$46 : GOSUB OPERATION // NOT
OP = OP_AND : A = \$41 : B = \$AB : GOSUB OPERATION // AND
OP = OP_AND : A = \$85 : B = \$96 : GOSUB OPERATION // AND
OP = OP_OR : A = \$AB : B = \$1F : GOSUB OPERATION // OR
OP = OP_OR : A = \$02 : B = \$64 : GOSUB OPERATION // OR
OP = OP_XOR : A = \$1A : B = \$2B : GOSUB OPERATION // XOR
OP = OP_XOR : A = \$3C : B = \$4D : GOSUB OPERATION // XOR
OP = OP_XOR : A = \$5E : B = \$5E : GOSUB OPERATION // XOR
OP = OP_ADD : A = \$AB : B = \$1F : GOSUB OPERATION // ADD
OP = OP_ADD : A = \$F2 : B = \$64 : GOSUB OPERATION // ADD
OP = OP_SUB : A = \$AB : B = \$53 : GOSUB OPERATION // SUB
OP = OP_SUB : A = \$68 : B = \$E7 : GOSUB OPERATION // SUB
OP = OP_SUB : A = \$1A : B = \$1A : GOSUB OPERATION // SUB
OP = OP_SHL : A = \$EA : GOSUB OPERATION // SHL
OP = OP_SHL : A = \$2B : GOSUB OPERATION // SHL
OP = OP_SHR : A = \$EA : GOSUB OPERATION // SHR
OP = OP_SHR : A = \$07 : GOSUB OPERATION // SHR
OP = OP_CMP : A = \$AB : B = \$53 : GOSUB OPERATION // CMP
OP = OP_CMP : A = \$68 : B = \$E7 : GOSUB OPERATION // CMP
OP = OP_CMP : A = \$1A : B = \$1A : GOSUB OPERATION // CMP
ENDIF

END

OPERATION:

MODE = (OP >> 7) & 1 : CIN = (OP >> 6) & 1
S3 = (OP >> 5) & 1 : S2 = (OP >> 4) & 1
S1 = (OP >> 3) & 1 : S0 = (OP >> 2) & 1
SHR = (OP >> 1) & 1 : OE = (OP >> 0) & 1
SLEEP FOR 1m
RETURN

*ENDSCRIPT

*SCRIPT GENERATOR TEST_ALU_CLOCK

ALIAS ONOFF = \$ON\$/OFF

PIN ONOFF
PIN CLK = FLT

INT i

IF ONOFF = TRUE THEN
CLK = FLT

ELSE

SLEEP FOR 0.5m
FOR i = 0 TO 30 STEP 1
SLEEP FOR 0.5m
CLK = 0
SLEEP FOR 0.5m
CLK = 1

NEXT i

ENDIF

END

*ENDSCRIPT

